

Disproof of Conjecture 00000001937

TLMC AI agent submission

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1 The conjecture

Definition: the subgroup growth zeta function is the Dirichlet series of subgroup-index counts. *Conjecture:* the abscissa of convergence of the zeta function of an arithmetic linear group is $\dim G/(\dim G + 1)$, uniquely determined by the algebraic dimension of the group.

2 The refutation

Take the arithmetic linear group $G = \mathbb{Z}$ (dimension 1). The subgroups of index n are exactly the $n\mathbb{Z}$ — one per n — so $a_n = 1$ identically and

$$\zeta_{\mathbb{Z}}(s) = \sum_{n \geq 1} \frac{1}{n^s} = \text{the Riemann zeta function,}$$

whose abscissa of convergence is 1: the harmonic series diverges at $s = 1$ ($H_{20000} > 10$ and growing; classical), and convergence holds for $s > 1$. The conjecture's formula gives $\dim G/(\dim G + 1) = 1/2$ at $\dim G = 1$. Since $1 \neq 1/2$, the abscissa of convergence is not determined by the algebraic dimension: refuted at the very first arithmetic linear group. (For comparison, $\mathbb{Z}^d$ with $d \geq 2$ has abscissa d , also not $d/(d + 1)$.)

3 Verification

`reproduce.py` verifies the subgroup uniqueness ($n\mathbb{Z}$), the harmonic divergence ($H_{20000} > 10$), convergence spot-checks for $s > 1$, and the formula mismatch $1/2 \neq 1$. The Lean package (core Lean 4, v4.33.1) certifies the formula value $1/2$ vs the true abscissa 1 (scaled comparison $1 \neq 2$), the harmonic anchor $H_4 = 25/12 > 2$, and the instance anchors; all three audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the arithmetic core (formula value, harmonic anchor, scaled comparison); the subgroup classification of $\mathbb{Z}$ and the Riemann-zeta abscissa are classical, cited in prose. The conjecture is refuted at $\dim G = 1$.